Harsh Mali

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EDUCATION

The Pennsylvania State University **GPA: 3.89/4.00**

Honors Bachelor of Science in Mechanical Engineering, University Park, PA *Aug. 2025 - May 2028*

RELEVANT COURSEWORK

- Dynamics
- Eng. Thermodynamics
- Strength of Materials
- Statics

EXPERIENCE

S.H.A.P.E. Lab Jan. 2026 – Present
Research Assistant *University Park, PA*
• Co-authored research on multi-material L-PBF of Cu10Sn bronze and 316L stainless steel, using in-situ synchrotron X-ray imaging and CFD simulations to study spatter, melt-pool instability, and material mixing.
• Developed Python-based image-processing workflows to analyze high-speed experimental imaging of metal additive manufacturing processes.
• Beginning thesis research integrating Python and CAD model analysis to characterize and predict the geometry of 3D-printed sand-casting molds.

American Society of Mechanical Engineers Aug. 2025 - Present
Project Lead & Prev. Manufacturing Lead *University Park, PA*
• Led manufacturing and assembly of a remotely operated waste-collection robot, meeting strict competition size, safety, and performance constraints.
• Contributed to robot design, fabrication, and testing, helping secure 3rd place nationally at the EFx Competition.
• Currently Overall Project Lead for this year's EFx Robotics Team and am responsible for project planning, technical development, manufacturing operations, and team coordination.

Nittany Rocket Labs Jan. 2026 - Present
Payload Member *University Park, PA*
• Designed and integrated an electronic payload system to collect, process, and store in-flight data for the International Rocket Engineering Competition (IREC).
• Developed and tested embedded software, sensor interfaces, and telemetry systems for real-time data acquisition and transmission to a ground station.
• Continuing to improve payload electronics and avionics through system integration, validation testing, and flight preparation for this year's IREC.

Undergraduate Research and Fellowships Mentoring April 2026 - Present
Mentor *University Park, PA*
• Guide students in securing research opportunities by providing mentorship on faculty outreach, application strategies, and campus resources.
• Advocate for undergraduate participation in research and fellowship programs through peer mentorship and engagement initiatives.

PROJECTS

Reaction Wheel Balance Controller| *Arduino, C++, IMU* Jan. 2026 – May 2026
• Developed a single-axis self-stabilizing platform using a reaction wheel and closed-loop feedback control to maintain autonomous balance.
• Designed and integrated mechanical, electrical, and embedded systems, including an Arduino, motor driver, and MPU6050 IMU, and validated performance through iterative testing and controller tuning.

TECHNICAL SKILLS

Languages: Python, MATLAB, C++, HTML

Engineering Software: Onshape, Fusion 360, Inventor, Autodesk CFD, N-Top, SolidWorks, Inventor FEA, Generative Design, Microsoft 365(Excel, Word, etc.), Robotic Design, Arduino, Circuitry